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GitHub: <https://github.com/FratresMedAI/Ancile-Aeris>

Ancile Aeris v2.0 Concept Paper

LRBA Counter-UAS / Anti-Terror Defensive System Concept Property of Fratres X AI GitHub:

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Why Ancile Aeris

Ancile Aeris v2.0 gives DHS evaluators a working, auditable C-UAS software layer that moves beyond detection into human-governed, layered non-kinetic defeat. It demonstrates the exact operational balance required for homeland defense: decisive effect selection, clean auditability, and no autonomous weapon release.

Executive Summary

Ancile Aeris v2.0 is a fully integrated ROS 2 counter-UAS (C-UAS) and anti-terror defensive platform for mass-gathering security, critical infrastructure defense, and modular federal acquisition pathways. The verified stack combines multi-modal sensing, deterministic fusion, mesh-coordinated scout mothership ISR, PID ≥ 0.999 safety gates, clean DARKSPACE audit status, cognitive EW recommendation logic, and a full layered non-kinetic defeat stack.

The central innovation is `ancile_aeris_effectors`: a simulation-safe effector selection layer modeling HPM-class denial, cognitive jamming, GNSS/link spoofing, authorized control-link takeover, multi-sensor deception, and monitor mode. The cognitive chain selects among these options and publishes explainable AI (XAI) rationale, producing a lethally effective but human-governed defensive architecture.

Final verification confirms a clean demonstration posture: `/darkspace/status` reports `integrity_ok: true` and `chain_gap_count: 0`; `/effector/selected_plan`, `/effector/status`, and `/cognitive_ew_commands` publish live; and the system demonstrates sensing-to-fusion-to-effectors-to-audit continuity.

Mission Need

Adversarial small UAS threats compress response timelines around stadiums, transit nodes, energy infrastructure, ports, public events, and national special security activities. Operators need more than detection: they need defensible decision support that can correlate evidence, recommend low-collateral defeat options, preserve human authority, and generate audit artifacts suitable for after-action review.

Ancile Aeris answers that need as a system-of-systems software layer: sensing, fusion, safety, mesh ISR, cognitive selection, non-kinetic effect modeling, XAI, and audit all operate together through ROS 2 topic contracts.

Technical Approach

Ancile Aeris v2.0 is organized as a modular ROS 2 workspace. The LRBA demonstration slice launches with:

```
ros2 launch ancile_aeris_bringup ancile_aeris_basic_demo.launch.py
```

Sensing, Fusion, and ISR

The platform simulates visual, thermal, acoustic, RF, lidar, and SIGINT-style inputs. These feed /fused_tracks, producing structured track records with confidence, modality coverage, and simulation-labeled counter-terror cues.

The scout_mothership package adds mesh-coordinated ISR through:

- ? /scout_eyes
- ? /mesh/mothership_swarm_status
- ? /mesh/mothership_peers/heartbeat

This supports an Anduril Lattice-style common operating picture: distributed coverage, sector assignment, resilient awareness, and defensive handoff logic.

Safety Gates and Human Authority

PID >= 0.999 safety context constrains downstream recommendations. High-consequence options remain monitor-only unless configured human approval paths are satisfied. Control-link takeover concepts require dual-authorization semantics. The demo separates recommendation, simulation, and actuation: it recommends layered defensive responses but never autonomously releases a weapon or drives real effector hardware.

DARKSPACE Audit

DARKSPACE subscribes to /audit/events and publishes /darkspace/status. The v2.0 audit fix isolates hash chains per emitting process using chain_id, preventing independent publishers from being evaluated as one chain. Final verification showed:

- ? integrity_ok: true
- ? chain_gap_count: 0
- ? zero darkspace hash-chain mismatch observed warnings

This gives evaluators a clean oversight narrative: each major event is observable, traceable, and suitable for compliance review.

Core Innovation: Layered Non-Kinetic Defeat

ancile_aeris_effectors is the v2.0 differentiator. It makes non-kinetic defeat visible, selectable, auditable, and explainable.

Effector Mode	Defensive Role	Safety Posture
monitor	Passive ISR and no-action state	Default safe state
multi_sensor_deception	Deception / sensor disagreement modeling	Human-gated simulation
cognitive_jamming	Adaptive RF denial recommendation	Human-gated simulation
gnss_link_spoofing	Navigation / link deception modeling	Human-gated simulation
hpm_denial_stub	HPM-class non-kinetic denial concept	Simulation-only
control_link_takeover	Authorized recovered-link takeover concept	Dual authorization required

The package publishes /effector/selected_plan and /effector/status. Each selected plan includes track ID, selected mode, family, score, authorization state, monitor-only status, catalog considered, and XAI rationale.

Cognitive Selection and XAI

Ancile Aeris v2.0 connects effectors into the cognitive EW path:

```
agent_orchestrator -> digital_twin -> cognitive_ew -> /cognitive_ew_commands
```

The cognitive output includes `selected_effector_mode`, `effector_family`, `effector_plan_id`, `xai_rationale`, `monitor_only`, and RF recommendation parameters. This proves the platform is not simply listing effects; it is selecting a defensible non-kinetic option and explaining why.

Government and Industry Alignment

Ancile Aeris v2.0 aligns with DHS and defense modernization priorities:

- ? **DHS Program Executive Office for UAS/C-UAS:** defensive C-UAS, event protection, and critical infrastructure security.
- ? **Anduril Lattice:** mesh ISR and common operating picture analogs.
- ? **JIATF-401 Marketplace:** modular acquisition and integration pathway.
- ? **Fortem DroneHunter F700-class:** low-collateral intercept/capture reference for optional simulation pathways.
- ? **Epirus Leonidas:** HPM-class non-kinetic effect framing represented as a safe software stub.
- ? **Replicator 2:** rapid software-defined defense fielding posture.

Simulation-only C-UAS defensive demonstration. No autonomous weapon release. Government and industry references are alignment examples only and do not imply endorsement.

Team and Execution Readiness

Fratres X AI has delivered a working, auditable ROS 2 defense-tech prototype with buildable packages across sensing, fusion, safety, audit, scout ISR, cognitive EW, bringup, and non-kinetic effectors. The repository is live, tagged, and verified with clean runtime outputs for DARKSPACE status, effector selection, cognitive EW, and topic-level proof points.

The architecture is intentionally modular: each capability exposes clear ROS 2 topics and JSON contracts, making Ancile Aeris extensible to future sensors, dashboards, decision models, and integration testbeds.

Impact

Ancile Aeris v2.0 gives DHS evaluators a concrete, inspectable software layer for next-generation defensive C-UAS:

- ? **Operational:** faster track correlation, mesh ISR, and layered response recommendations.
- ? **Technical:** non-kinetic effectors are visible, selectable, auditable, and explainable.
- ? **Oversight:** DARKSPACE status and XAI outputs support review and legal defensibility.
- ? **Acquisition:** the stack maps cleanly to modular marketplace and rapid-fielding pathways.

Ancile Aeris v2.0 demonstrates a C-UAS system that is lethally effective in defeat logic while remaining human-governed, simulation-safe, and submission-credible.

Production Notes

- ? Convert to PDF with Fratres X AI branding, repository link, and one architecture figure.
- ? Recommended visuals: ROS 2 topic graph, effector catalog, DARKSPACE status screenshot, and mesh ISR diagram.
- ? Use the standard disclaimer verbatim in the footer or final page.

Ancile Aeris v2.0 is ready for evaluation, integration discussions, and acquisition pathway alignment.

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